CSE400: Fundamentals of Probability in Computing March 24, 2026

Inequalities and Tail Bounds

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1 Introduction to Tail Bounds

While the mean provides the average behavior of a random variable, tail bounds allow us to quantify the probability of "rare extreme events"[cite: 1391]. These are defined by the tail of a random variable X: $P\{X > x\}$[cite: 1373]. Tail bounds are essential in computing for analyzing scenarios like jobs delayed over 24 hours or packets dropped due to full buffers[cite: 1375].

2 Chebyshev's Inequality

Chebyshev's Inequality is more powerful than Markov's Inequality because it utilizes both the **mean** and the **variance** to control the deviation from the mean[cite: 1806, 1822].

2.1 Definition

For any random variable X with mean μ and variance σ^2, and for any $a > 0$:

$$P(|X-\mu| \geq a) \leq \frac{Var(X)}{a^2} \tag{1}$$

[cite: 1825, 1846].

2.2 Proof of Chebyshev's Inequality

The proof relies on applying Markov's Inequality to the non-negative random variable $(X-\mu)^2$[cite: 1847]:

1. Start with the tail probability: $P(|X-\mu| \geq a)$.
2. Square both sides of the inequality: $P((X-\mu)^2 \geq a^2)$[cite: 1848].
3. Apply Markov's Inequality ($P(Y \geq k) \leq \frac{E[Y]}{k}$):

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2} \tag{2}$$

4. By definition, $E[(X-\mu)^2] = Var(X)$, thus:

$$P(|X-\mu| \geq a) \leq \frac{Var(X)}{a^2} \tag{3}$$

[cite: 1849].

2.3 Solved Example: Exam Grades

Problem: Average grade $\mu = 70$, standard deviation $\sigma = 5$. Find the bound for the fraction of students receiving an "A" (cutoff 90)[cite: 1930].

- **Step 1:** Identify $\mu = 70$ and $\sigma^2 = 25$[cite: 1934].
- **Step 2:** $X \geq 90 \implies |X - 70| \geq 20$[cite: 1945].
- **Step 3:** $P(|X - 70| \geq 20) \leq \frac{25}{20^2} = \frac{25}{400} = \frac{1}{16}$[cite: 1972, 1988].
- **Result:** At most 6.25% of students receive an "A", which is much tighter than the Markov bound of 77.8%[cite: 1985, 1988].

3 Chernoff Bound

The Chernoff bound provides the tightest tail bounds by using the **Moment Generating Function** (e^{tX})[cite: 2071, 2122].

3.1 Core Trick and Form

The transformation magnifies large values of X exponentially to create a tighter bound[cite: 2079, 2108].

$$P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}} \text{ for } t > 0 \tag{4}$$

[cite: 2162, 2170]. The final form is minimized over t:

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\} \tag{5}$$

[cite: 2186].

3.2 Poisson Tail Bound Derivation

For $X \sim Poisson(\lambda)$, the MGF is $E[e^{tX}] = e^{\lambda(e^t - 1)}$[cite: 2246]. Setting $t = \ln(a/\lambda)$ to minimize the exponent[cite: 2259]:

$$P(X \geq a) \leq e^{a-\lambda} \left(\frac{\lambda}{a} \right)^a \tag{6}$$

[cite: 2270].

4 Jensen's Inequality

Jensen's Inequality relates the value of a convex function of an integral to the integral of the convex function[cite: 2544].

4.1 Convex Functions

A function $g(x)$ is convex if $g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha)g(x_2)$[cite: 2440]. Mathematically, $g''(x) \geq 0$[cite: 2461].

4.2 Theorem

If $g(x)$ is convex on an interval S and X takes values in S, then:

$$g(E[X]) \leq E[g(X)] \tag{7}$$

[cite: 2546]. Interpretation: Averaging first gives a smaller value than applying the function first[cite: 2555, 2556].

5 Summary for Review

Inequality	**Information Used**	**Bound Type**	**Tightness**
Markov	Mean	Polynomial ($1/a$)	Loose [cite: 1622]
Chebyshev	Mean + Variance	Polynomial ($1/a^2$)	Moderate [cite: 1872]
Chernoff	Full Distribution (MGF)	Exponential (e^{-a})	Strongest [cite: 2204]

Golden Rule: More statistical information leads to tighter bounds, which results in lower hardware costs in system design.